MILITARY SPECIFICATION

TRANSMITTER, RELAY, MARK 44 MOD 1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the requirements for the procurement of Relay Transmitter Mk 44 Mod 1, BuWeps LD 534037. The relay transmitter receives missile-in-flight position information in computed polar coordinates. The purpose of the relay transmitter is to provide bi-directional compatability between an attack console and either missile or gunfire control radar systems. This relay transmitter is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents or such portions thereof as may be designated herein, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-plywood

Military

MIL-P-116	Preservation, Methods of
MIL-A-5092	Adhesive, Rubber (Synthetic and Reclaimed Rubber Base)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-G-12803	Gasket, Materials, Nonmetallic

FSC 1265

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MIL-T-17113	Tests; Shock, Vibration, and Inclination (For Electronic Equipment) General Specification
MIL-I-26860	Indicator, Humidity, Plug, Color Change

Bureau of Naval Weapons

OS 6341	Miscellaneous General Ordnance Design Requirements

STANDARDS

Military

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for shipment and Storage
MIL-STD-167 (SHIPS)	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

Bureau of Naval Weapons

LD 534037	Relay Transmitter Mk 44 Mod 1 Assembly
1815417	Container, Shipping, Assembly

Bureau of Ships

807-655947	High-Shock Testing Machine, Medium-Weight

PUBLICATIONS

Bureau of Naval Weapons

OP 1700	Standard Fire Control Symbols
OD 13808	Relay Transmitter Mk 44 Mod 1, Factory Acceptance Test

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 General.

3.1.1 Materials and treatments of materials and parts. Unless otherwise specified, all materials, treatments, and parts used in the manufacture of Relay Transmitter Mk 44 Mod 1 shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 534037.

3.1.2 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in BuWeps LD 534037.

3.1.3 Gages. The contractor shall provide whatever gages are necessary and adequate to ensure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.1.4 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and all supplements thereto. The class of fit for threads shall be as specified on the applicable drawings, and the gaging practices shall be as set forth in the supplement to the aforementioned Handbook H28.

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3.1.5 Inspection and test facilities. The supplier shall furnish and maintain all necessary inspection and test facilities, and shall provide personnel for performing all acceptance tests. The test equipment shall be accurate enough to ensure the validity of the test results.

3.1.6 Finishes, coatings, construction, and dimensions. Relay Transmitter Mk 44 Mod 1 finishes, coatings, construction, and dimensions shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BuWeps LD 534037.

3.1.7 Moisture, fungus, and corrosion. Relay Transmitter Mk 44 Mod 0 shall be rendered moisture-, fungus-, and corrosion-resistant in accordance with detailed requirements specified on the drawings and specifications listed in BuWeps LD 534037.

3.2 Performance and product characteristics.

3.2.1 Performance.

3.2.1.1 Synchro rotation. With increasing functions transmitted to or from the relay transmitter, the synchro shafts shall rotate counter-clockwise and the dials shall rotate clockwise.

3.2.1.2 Limit stops. Dials limited by mechanical stops shall indicate the values specified in table I when the stops are engaged.

Table I. Limit Stops

Function	Limit	Tolerance
Designated Missile Elevation, d(Em)	2900 minutes	Plus 0 minus 20 minutes (arc)
	to	
	5900 minutes	Plus 20 minus 0 minutes (arc)
Effective Range, Rh5	0 yards	Plus 0 minus 50 yards
	to	
	99,000 yards	Plus 50 minus 0 yards
Time to Separation, T4	0 seconds	Plus 0 minus 1 seconds (time)
	to	
	55 seconds	Plus 1 minus 0 seconds (time)

3.2.1.3 Synchro zero settings. All synchro zero settings shall be as specified in table II. All synchros are 400-cycle, except as noted in the table.

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Table II. Synchro Zero Settings

Synchro	Function	Speed	Zero Setting
8A1B3	T4	360 sec/rev (time)	10 sec
8A1B4	T4	10 sec/rev (time)	10 sec
8A2B2	Rh5	1,296,000 yd/rev	10,000 yd
8A2B3	Rh5	72,000 yd/rev	10,000 yd
8A2B4	Rh5	2,000 yd/rev	10,000 yd
8A2B5*	d(Rm)	72,000 yd/rev	10,000 yd
8A3B3†	By5	360 deg/rev	0 deg
8A3B4†	By5	10 deg/rev	0 deg
8A7B3	d(Em)	180 deg/rev	2000 min
8A7B4	d(Em)	10 deg/rev	2000 min
8A7B5	d(Em)	360 deg/rev	2000 min
8A7B6*	d(Em)	180 deg/rev	2000 min
8A8B3	Bm	360 deg/rev	0 deg
8A8B4	Bm	10 deg/rev	0 deg
8A8B5	Bm	360 deg/rev	0 deg
8A9B3	d(Bmy)	360 deg/rev	0 deg
8A9B4	d(Bmy)	10 deg/rev	0 deg

* 60-cycle synchro.

† Used in True Water-Entry Point Bearing, By5, servo module, if installed.

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3.2.1.4 Synchro inputs. Relay Transmitter Mk 44 Mod 1 shall be capable of receiving the input quantities specified in table III.

Table III. Synchro Inputs

Quantity	Range	Allowable Error
By5*	0 to 360 deg	0.1 degree
T4	0 to 55 sec	0.1 second (time)
Rh5	0 to 99,000 yd	5 yards
Bm	0 to 360 deg	0.5 degree
d(Bmy)	0 to 360 deg	0.1 degree

* Module provided for but not installed.

3.2.1.5 Synchro outputs. Relay Transmitter Mk 44 Mod 1 shall be capable of transmitting synchro output quantities as specified in table IV.

Table IV. Synchro Outputs

Quantity	Range	Allowable Error
d(Em)*	2900 to 5900 min	0.3 degree
d(Em)	2900 to 5900 min	0.3 degree
d(Em)	2900 to 5900 min	0.3 degree
Bm*	0 to 360 deg	0.3 degree
d(Rm)	0 to 33,000 yd	0.3 degree

* Two-speed outputs.

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3.2.1.6 Scaled voltage outputs. Relay Transmitter Mk 44 Mod 1 shall be capable of transmitting scaled output quantities as specified in table V.

Table V. Scaled Outputs

Quantity	Voltage	Allowable Error	Range
d(Em)	0.00222 volt per minute of elevation	±1%	2,900 to 5,900 min
d(Rmx)*	1 volt per 10,000 yd	±1%	0 to 33,000 yd
d(Rmy)*	1 volt per 10,000 yd	±1%	0 to 33,000 yd
Rh5x	5 volts per 10,000 yd	±1%	0 to 99,000 yd
Rh5y	5 volts per 10,000 yd	±1%	0 to 99,000 yd

* Two separate outputs of the quantity shall be available

3.2.1.7 Pulsed voltage outputs. Relay Transmitter Mk 44 Mod 1 shall be capable of transmitting pulsed voltage output quantities as specified in table VI.

Table VI. Pulsed Voltage Outputs

Quantity	Characteristics	Allowable Error
T4-T	Start 3 seconds prior to separation of missile and airframe. (Pulse duration 3 seconds)	±1 sec

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3.2.2 Product characteristics.

3.2.2.1 Indicator lamps. The relay transmitter indicator lamps shall go on only when handknobs are engaged and shall go off only when handknobs are disengaged.

3.2.2.2 Dielectric strength. The synchro and servomotor circuits of Relay Transmitter Mk 44 Mod 1 shall be capable of withstanding 700 volts AC, rms, 60 cps, applied to pins of the external connectors and ground (case). The low-voltage relay, amplifier, and illumination circuits shall be capable of withstanding 325 volts AC, rms, 60 cps, from pins of the external connectors and ground (case).

3.2.2.3 Insulation resistance. The circuits of Relay Transmitter Mk 44 Mod 1 shall have a minimum of 3 megohms insulation resistance from terminals of the external connectors to ground (case).

3.2.2.4 Degree of enclosure. The relay transmitter shall be enclosed in a spraytight case in accordance with the requirements specified in Military Standard MIL-STD-108, Spraytight.

3.3 Environmental requirements.

3.3.1 Vibration. Relay Transmitter Mk 44 Mod 1 shall be capable of withstanding Type I vibration as specified in Military Standard MIL-STD-167.

3.3.2 Shock. Relay Transmitter Mk 44 Mod 1 shall meet the following shock requirements:

3.3.2.1 Shock, transportation, storage, and handling. Relay Transmitter Mk 44 Mod 1, packed and packaged as specified in 5.1 and 5.2, shall be capable of withstanding shock encountered in transportation, storage, and handling in accordance with the Free-Fall Drop Test of Specification MIL-P-116C.

3.3.2.2 Shock, operating condition. Relay Transmitter Mk 44 Mod 1 shall be capable of withstanding shock encountered in shipboard environment in accordance with Specification MIL-T-17113.

3.3.3 Inclination. Relay Transmitter Mk 44 Mod 1 shall be capable of withstanding pitch and roll motion encountered in shipboard environment in accordance with Specification MIL-T-17113.

3.3.4 Temperature and humidity. Relay Transmitter Mk 44 Mod 1 shall be capable of withstanding temperature and humidity encountered in shipboard environment in accordance with NAVORD OS 6341.

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3.3.5 Fungus resistance. Relay Transmitter Mk 44 Mod 1 shall be capable of resisting fungus growth in accordance with Specification MIL-E-5272.

3.4 Clean-up. Prior to and after final assembly, all components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.5 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts, over-all assembly, painting, and machine screw assemblage. All parts shall be free of burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Government inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Production sample. At the option of the contracting agency, a Relay Transmitter Mk 44 Mod 1 may be selected by the Government inspector for reliability testing as outlined in 4.5 to determine compliance of the sample with the requirements of the contract, specification, and drawings.

4.2 Classification of tests. The inspection and testing of Relay Transmitter Mk 44 Mod 1 shall be classified as follows:

4.2.1 Acceptance tests. Acceptance tests shall be accomplished on all Relay Transmitters Mk 44 Mod 1 being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production.

4.2.2 Tests after delivery. Unless otherwise specified in the contract, the Government reserves the right to conduct any or all tests specified herein as deemed necessary after shipboard installation of the relay transmitter. Any defects discovered during or after these tests due to the contractors failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

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4.2.3 Reliability tests. Reliability tests are those which are accomplished on a sample selected which is representative of the production of Relay Transmitter Mk 44 Mod 1 after the award of the contract, to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on absence of defects in a sample. Failure in any of these tests will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.3 Test equipment. The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 Voltage breakdown tester. A voltage breakdown tester having voltage ranges of 0 to 1000 volts (rms) AC and 0 to 600 volts DC, with maximum currents specified at 7 milliamperes AC and 5 milliamperes DC, respectively.

4.3.2 Megohmer. A megohmer having a test potential of 500 volts DC and ranges of 0 to 200 megohms and 0 to 2000 ohms.

4.3.3 Voltmeters. A DC microvoltmeter capable of measuring from microvolts to 100 volts. An AC voltmeter capable of measuring from 1-millivolt to 300 volts.

4.3.4 Dummy synchro transmitters. The required dummy synchro transmitters shall be as specified in table VII.

Table VII. Dummy Synchro Transmitter Requirements

Speed	Ratio	Frequency (cps)	Dial Graduation	Function	Electrical Accuracy
Single	--	400	Degrees	Bm	±0.6 min
Double	36:1	400	Degrees	T4	±0.6 min
Double	36:1	400	Degrees	d(Bmy)	±0.6 min
Triple	648:36:1	400	Degrees	Rh5	±0.6 min

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4.3.5 Dummy synchro receivers. The required dummy synchro receivers shall be as specified in table VIII.

Table VIII. Dummy Synchro Receiver Requirements

Speed	Ratio	Frequency (cps)	Dial Graduation	Function	Electrical Accuracy
Single	--	60	Degrees	d(Em)	±12 min
Single	--	400	Degrees	d(Em)	±12 min
Single	--	400	Degrees	Bm	±12 min
Single	--	60	Degrees	d(Rm)	±12 min
Single	--	400	Degrees	d(Em)	±12 min

4.3.6 Stop watch. A stop watch having a capability of measuring a 3-minute period with accuracy of 0.1 second.

4.3.7 Decade voltage divider. A decade voltage divider having a range of 0.0000 to 1.0000 in steps of 0.001 and an accuracy of plus or minus 0.1 percent.

4.4 Acceptance tests. The contractor or Government Inspector shall conduct acceptance tests as specified in 4.2.1 to assure that the relay transmitter complies with the requirements of 3.2, 3.3, 3.4 and 3.5.

4.4.1 Test conditions. Unless otherwise specified, the relay transmitter shall be subjected to acceptance tests under the following conditions:

4.4.1.1 Temperature: Room ambient 65 degrees F. to 110 degrees F. (18.33 degrees C. to 44.66 degrees C.).

4.4.1.2 Altitude. Normal ground atmospheric pressure.

4.4.1.3 Vibration. None.

4.4.1.4 Humidity. Room ambient to 95 percent relative maximum.

4.4.2 Test procedures.

4.4.2.1 Synchro rotation tests. When tested in accordance with the Static Transmission Test Problems of NAVWEPS OD 13808, all dials shall meet the requirements of 3.2.1.1 of this specification for synchro rotation.

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4.4.2.2 *Limit stop tests.* When the relay transmitter is connected and tested in accordance with the Limit Stop Adjustment Check of NAVWEPS OD 13808, the limit stops shall meet the adjustment requirements specified in table I of this specification.

4.4.2.3 Synchro zero setting tests. When the synchros are tested in accordance with the Synchro Electrical Zero Check of NAVWEPS OD 13808, all synchro zero settings shall be as specified in table II of this specification.

4.4.2.4 Indicator lamp tests. When the relay transmitter is connected and tested in accordance with the Indicator Lamp Check of NAVWEPS OD 13808, the indicator lamp on each module shall go on only when each handknob is engaged, and shall go off only when the handknob is disengaged.

4.4.2.5 Input tests.

4.4.2.5.1 Time to separation. When connected and tested in accordance with the Static Input Transmission Test of NAVWEPS OD 13808, the relay transmitter shall automatically receive time to separation, T4, input values of 0 to 55 seconds.

4.4.2.5.2 Designated missile elevation. When connected and tested in accordance with the Static Input Transmission Test of NAVWEPS OD 13808, the relay transmitter shall receive designated missile elevation, d(Em), manual input values of 2900 and 5900 minutes.

4.4.2.5.3 Effective range. When connected and tested in accordance with the Static Input Transmission Test of NAVWEPS OD 13808, the relay transmitter shall automatically receive effective range, Rh5, input values of 0 to 99,000 yards.

4.4.2.5.4 Relative missile bearing. When connected and tested in accordance with the Static Input Transmission Test of NAVWEPS OD 13808, the relay transmitter shall automatically receive relative missile bearing, Bm, input values of 0 to 360 degrees.

4.4.2.5.5 Designated true missile bearing. When connected and tested in accordance with the Static Input Transmission Test of NAVWEPS OD 13808, the relay transmitter shall automatically receive designated true missile bearing, d(Bmy), input values of 0 to 360 degrees.

4.4.2.5.6 True water-entry point bearing. When connected and tested in accordance with the Static Input Transmission Test of NAVWEPS OD 13808, the relay transmitter shall automatically receive true water-entry point bearing, By5, input values of 0 to 360 degrees.

4.4.2.6 Output tests.

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4.4.2.6.1 Time to separation. When connected and tested in accordance with the Static Output Transmission Test of NAVWEPS OD 13808, the relay transmitter shall transmit time to separation, T4, output values as signal voltages, T4-T.

4.4.2.6.2 Designated missile elevation. When connected and tested in accordance with the Static Output Transmission Test of NAVWEPS OD 13808, the relay transmitter shall transmit designated missile elevation, d(Em), output values as scaled voltages and synchro signals.

4.4.2.6.3 Designated missile range. When connected and tested in accordance with the Static Output Transmission Test of NAVWEPS OD 13808, the relay transmitter shall transmit designated missile range, d(Rm), as scaled voltages and synchro signals.

4.4.2.6.4 Effective range. When connected and tested in accordance with the Static Output Transmission Test of NAVWEPS OD 13808, the relay transmitter shall transmit effective range, Rh5, output values as scaled voltages.

4.4.2.6.5 Relative missile bearing. When connected and tested in accordance with the Static Output Transmission Test of NAVWEPS OD 13808, the relay transmitter shall transmit relative missile bearing, Bm, synchro output values of 0 to 360 degrees.

4.4.2.7 Dielectric strength tests. When connected and tested in accordance with the Dielectric Test of NAVWEPS OD 13808, the relay transmitter shall meet the requirements of this specification.

4.4.2.8 Insulation resistance test. When connected and tested in accordance with the Insulation Resistance Test of NAVWEPS OD 13808, the relay transmitter shall meet the requirements of this specification.

4.4.2.9 Degree of enclosure test. The relay transmitter shall be subjected to the spraytight enclosure test specified in Military Standard MIL-STD-108, Spraytight.

4.5 Reliability tests.

4.5.1 Vibration.

4.5.1.1 Vibration, operating condition.

4.5.1.1.1 Method of attachment. For all tests, the relay transmitter shall be secured to the mounting bracket of the testing machine in the same manner that it shall be secured on board ship.

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4.5.1.1.1.1 Orientation for vibration test. The relay transmitter shall be installed on vibration testing machine in such a manner that the direction of vibration will be in turn along each of the three mutually perpendicular axes of the instrument. On a horizontal vibration testing machine, the relay transmitter may be turned 90 degrees in the horizontal plane in order to vibrate it in each of the two horizontal orientations. The relay transmitter shall not be attached in any position other than those specified.

4.5.1.1.2 Vibration tests. Each of the tests outlined herein shall be conducted separately in each of the three principal directions of vibration. All tests in one direction shall be completed before proceeding with tests in another direction. The relay transmitter shall be secured to the vibration table as specified in 4.5.1.1.1 and shall be energized to perform its normal functions. All amplitudes of vibration specified herein are average amplitudes of the vibration table. If it is found necessary to repair any damage severe enough to constitute a failure of the equipment, the entire test shall be repeated unless otherwise directed by the bureau or agency concerned.

4.5.1.1.2.1 Exploratory vibration test. To determine the presence of resonances in the relay transmitter under test, the equipment shall be secured to the vibration table and vibrated at frequencies ranging from 5 cps to 33 cps at a table vibratory amplitude of 0.010 inch (plus or minus 0.002 inch). The change in frequency shall be made in discrete frequency intervals of 1 cps and maintained at each frequency for about 15 seconds. The frequencies at which resonances occur shall be noted.

4.5.1.1.2.2 Variable frequency test. The relay transmitter shall be vibrated from 5 cps to 33 cps in discrete frequency intervals of 1 cps at the amplitudes shown in table IX. At each integral frequency, the vibration shall be maintained for 5 minutes.

4.5.1.1.2.3 Endurance test. The relay transmitter shall be vibrated for a total period of at least 2 hours at the resonant frequencies chosen by the test engineer. If no resonance was observed, this test shall be performed at 33 cps. The amplitudes of vibration shall be in accordance with table IX.

Table IX. Amplitudes of Vibration

Frequency range (cps)	Table amplitude (inch)
5 to 15	0.030 ± 0.006
16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002

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4.5.2 Shock.

4.5.2.1 Shock, transportation, storage, and handling. Relay Transmitter Mk 44 Mod 1, packed and packaged as defined in 5.1 and 5.2, shall be dropped cornerwise from a height of 30 inches onto a solid steel or concrete surface in such a manner that the corner of the package absorbs the full force of the fall. This test shall be repeated until each of the eight corners of the package has received a fall. The height of 30 inches refers to the distance from the steel or concrete surface to the nearest corner of the package when suspended prior to the fall. The fall shall be a free fall in that no ropes or other suspending apparatus shall be attached to the package during the fall.

4.5.2.2 Shock, operating condition. Relay Transmitter Mk 44 Mod 1 shall be subjected to the following shock tests:

4.5.2.2.1 Method of mounting. For all tests, the relay transmitter shall be secured to the mounting bracket of the test machine in the same manner that it will be secured on board ship.

4.5.2.2.2 Shock test. The shock test shall consist of six blows applied in three groups of two blows each. The height of the hammer drop shall be 0.75 foot for group one and 1.75 feet for groups two and three. The initial up-travel of the anvil table shall be 3 inches for groups one and two, and 1-1/2 inches for group three. The weight of the hammer shall be 3000 pounds. The shock testing machine shall comply with BuShips Dwg 807-655947.

4.5.3 Inclination. Relay Transmitter Mk 44 Mod 1 shall be subjected to the following inclination tests:

4.5.3.1 Method of mounting. For all tests, the relay transmitter shall be secured to the mounting bracket of the test machine in the same manner that it will be secured on board ship.

4.5.3.2 Inclination test.

4.5.3.2.1 The relay transmitter shall be inclined at the rate of five to seven cycles per minute in one plane to an angle of 45 degrees on either side of the vertical for a period sufficiently long to determine its electrical characteristics under such motion or for a minimum of 30 minutes.

4.5.3.2.2 The test specified in 4.5.3.2.1 shall be repeated with the equipment reoriented 90 degrees to the plane in which it was originally tested.

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4.5.3.2.3 At the conclusion of the cyclic tests specified in 4.5.3.2.1 and 4.5.3.2.2, the cyclic motion may be stopped and the inclination adjusted to any angle or angles within 45 degrees of the vertical for a sufficient time to investigate the electrical characteristics of the equipment.

4.5.4 Temperature and humidity. Relay Transmitter Mk 44 Mod 1 shall be subjected to the following test. Unless otherwise specified, temperature and humidity tolerances shall be ±2 degrees C. (±3.6 degrees F.) and ±5 percent, respectively. The relay transmitter shall be placed in the test chamber to simulate installed conditions. The temperature and relative humidity in the chamber shall be 49 degrees C. (120 degrees F.) and 95 percent, respectively. Test conditions shall be maintained for 360 hours. At the conclusion of this period, the relay transmitter shall be operated, the results recorded, and a visual examination shall be made for evidence of deterioration or corrosion.

4.5.5 Fungus resistance tests.

4.5.5.1 Procedure. Five species of fungi are listed in 4.5.5.1.1; one specimen of fungus from each group shall be used. In the preparation of the spore suspension, distilled water having a pH value between 5.8 and 7.2 at temperatures between 72 degrees F. and 89 degrees F. shall be used. Approximately 10 milliliters (ml) of distilled water shall then be introduced directly into a tube culture of the fungus, and the spores brought into suspension by vigorous shaking or by gentle rubbing of the spore layer with an inoculating loop without disturbing the agar surface. This process shall be repeated for each species of fungus. The separate spore suspensions that form the five species of fungi shall be mixed together to provide a composite suspension. Actively sporulating cultures between 7 and 21 days old after initial inoculation shall be used for the preparation of the spore suspension. After preparation, the spore suspension will be kept for not more than a 24-hour period at temperatures between 72 degrees F. and 89 degrees F., and not more than 48 hours at refrigerator temperatures of 35 degrees F. and 45 degrees F. The relay transmitter, including applicable external connections, shall be placed in the chamber, maintaining an internal temperature of 30 degrees C. ±2 degrees C. (86 degrees F. ±3.6 degrees F.) and a relative humidity of 95 percent (±5 percent), and sprayed with the suspension of mixed spores. The test period shall be 28 days. At the end of the test period, the equipment shall be examined visually for evidence of deterioration or corrosion.

4.5.5.1.1 Organisms.

Group I	Chaetomium globosum USDA 1042.4 or Myrothecium verrucaria USDA 1334.2
Group II	Rhizopus nigricans S.N. 32 or Aspergillus niger USDA Tc 215-4247

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Group III	Aspergillus flavus WADC No. 26 or Aspergillus terreus PQMD 82J
Group IV	Penicillium luteum USDA 1336.1, Penicillium sp. USDA 1336.2 or Penicillium citrinum ATCC 9849
Group V	Memnoniella echinata WADC No. 37 or Fusarium moniliforme USDA 1004.1

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying, and preservative application. Not applicable.

5.1.1.2 Unit packaging. The relay transmitter shall be packaged in accordance with Method IIa of Specification MIL-P-116. Movable parts, susceptible to damage, shall be secured, blocked, and braced to prevent movement prior to packaging. Material for gaskets used to fabricate the package shall conform to Specification MIL-G-12803, Type 1, Class 2, Group 1, Grade 1. The adhesive used for cementing the gaskets shall conform to Specification MIL-A-5092 Type III. The humidity indicator for the package shall conform to Specification MIL-I-26860, Type II.

5.1.2 Level C

5.1.2.1 Cleaning, drying, and preservative application. Not applicable.

5.1.2.2 Unit packaging. The relay transmitter shall be packaged for domestic shipment in accordance with the manufacturer's commercial practice. The packaging shall be adequate to provide protection from deterioration and physical damage during direct shipment from source of supply to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Levels A and B. The relay transmitter, packaged as described in paragraph 5.1, shall be individually packed in a box, wood, cleated-plywood, conforming to Specification PPP-B-601, as specified in BuWeps Dwg 1815417. The relay transmitter shall be cushioned, blocked, and braced to withstand the shock tests for transportation, storage and handling of Section 4. The box shall be provided with an inspection port for viewing the humidity indicator.

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5.2.2 Level C. The transmitter, packaged as described in paragraph 5.1, shall be individually packed for domestic shipment in accordance with the manufacturer's commercial practice. The packing shall be adequate to provide protection from damage during direct shipment from source of supply to the first receiving activity for immediate use. The containers shall be in accordance with Uniform Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL-STD-129, including the serial number of the relay transmitter.

5.3.1.1 Special marking. The following special marking shall appear on the top, one side and one end of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES

6.1 Description. The relay transmitter is an electromechanical fire control instrument. It converts missile designation information received from an associated attack console to a form acceptable for use by weapon direction equipment for assignment to various missile or gunfire control radar systems. The relay transmitter also receives missile or target tracking information from various radar systems and converts this information to a form acceptable for use by an attack console. The relay transmitter consists of six servo modules enclosed in a spraytight case. External electrical connections are made through three cable connectors at the top of the case. The full-time services of an operator are not required because the instrument operates automatically. The relay transmitter is designed to operate satisfactorily for 3000 hours or a five year period, whichever occurs first.

6.2 Intended use. Relay Transmitter Mk 44 Mod 1 is intended for use in Fire Control System Mk 114 and Mods.

6.3 Ordering data. Procurement documents should specify the specific title, number, and date of this specification and exceptions to this specification, applicable drawings, and other documents.

6.4 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

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For Level B - This level shall be used only when it is definitely known that the item will be held in covered storage, either in domestic or overseas locations, for a period of six months or less.

For Level C - This level shall be used only when it is definitely known that the item is to be shipped to domestic installations for immediate use at the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level, and the contractor guarantees safe delivery.

6.5 Definitions.

6.5.1 Fire control symbols. For definitions of fire control symbols used throughout this specification, refer to OP 1700, Standard Fire Control Symbols.

6.5.2 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitube refers to displacement, which is given in inches, peak to peak. The frequency is given in cycles per second.

6.5.3 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock in general can be described as a build-up of 90 percent of peak force in less than 0.010 second. Shock is described as a pulse in units of gravitational acceleration (g's), duration in milliseconds (msec) and some description of the shock wave form.

6.5.4 Inclination. Inclination as referenced in this specification is defined as the acute angle between a given plane and the true vertical plane.

6.5.5 Tropical environment. Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which present an opportunity for fungi growth.

6.6 Power requirements. The relay transmitter shall have the power requirements shown in table X. The following legend provides a key to the abbreviations used in the table:

V - Nominal voltage rating in volts

F - Nominal frequency rating in cycles per second

P - Phase

I - Nominal current in amperes

KVA - Kilovolt-amperes requirement

PF - Load power factor

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KW - Kilowatt power requirement

NV - Maximum permissible nominal voltage variation in percent of rated voltage above and below rated voltage

NF - Maximum permissible nominal frequency variations in percent of rated frequency above and below rated frequency

NA - Not applicable

Table X. Power Requirements

Use	V	F (cps)	P (Ø)	I	KVA	PF	KW	NV (±%)	NF (±%)
Servo Module Power Supply	115	400	1	2.3	0.260	0.50	0.130	5.0	5
Synchro Transmitter	115	60	1	0.5	0.058	0.14	0.008	5.0	5
Illumination	28	60	1	1.4	0.039	1.00	0.039	10.0	10
Relays	28	0	NA	0.2	0.005	NA	0.005	10.0	NA
Signal Reference Voltage	60	0	NA	1.6	0.096	NA	0.096	0.5	NA
Signal Reference Voltage	12	400	1	10	0.120	1.00	0.120	0.5	6

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Governmental procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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